

Chapter 4, Section 3

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1. If X is a curve of genus 2, show that a divisor D is very ample $\iff \deg D \geq 5$. This strengthens (3.3.4).

Proof. Let D be a very ample divisor on X . By the degree-genus formula, a nonsingular plane curve cannot have genus equal to 2, and since D gives an embedding into projective space $|D|$, we must have $\ell(D) = \dim |D| + 1 \geq 4$. By Riemann-Roch, we have

$$\deg D = \ell(D) - \ell(K - D) + 1,$$

so it remains to show $\ell(K - D) = 0$. By (Ex. 1.5), $\deg D \geq 3$, so $\deg(K - D) < 0$. Hence, $\ell(K - D) = 0$. $\square$

3. If X is a curve of genus ≥ 2 which is a complete intersection (II, Ex. 8.4) in some $\mathbf{P}^n$, show that the canonical divisor K is very ample. Conclude that a curve of genus 2 can never be a complete intersection in any $\mathbf{P}^n$.

Proof. Let d be the degree of X in $\mathbf{P}^n$. Then X is a complete intersection of $n - 1$ hypersurfaces of degree d_i , and by (II, Ex. 8.4), we have $\mathcal{O}_X(K) \cong \mathcal{O}_X(\ell)$, where $\ell = \sum d_i - n - 1$. Thus, we have two ways of computing $\deg K$, and $g \geq 2$, so we have

$$\ell d = \deg K = 2g - 2 \geq 2,$$

which implies $\ell \geq 2$. Hence, $\mathcal{O}_X(K)$ is very ample. If X is a curve of genus 2, then $\ell(K) = 2$, so K corresponds to a morphism $X \rightarrow \mathbf{P}^1$, which can never be an embedding. $\square$

6. Curves of Degree 4.

- (a) If X is a curve of degree 4 in some $\mathbf{P}^n$, show that either

- (1) $g = 0$, in which case X is either the rational normal quartic in $\mathbf{P}^4$ (Ex. 3.4) or the rational quartic curve in $\mathbf{P}^3$ (II, 7.8.6), or
- (2) $X \subseteq \mathbf{P}^2$, in which case $g = 3$, or
- (3) $X \subseteq \mathbf{P}^3$ and $g = 1$.

- (b) In the case $g = 1$, show that X is a complete intersection of two irreducible quadric surfaces in $\mathbf{P}^3$ (I, Ex. 5.1).

Proof.

- (a) Assume X is not contained in any hyperplane of $\mathbf{P}^n$. An embedding $X \rightarrow \mathbf{P}^n$ is given by a linear system $\mathfrak{d}$ of degree d and dimension n , where there is a one-to-one correspondence between elements of $\mathfrak{d}$ and hyperplane sections of X in $\mathbf{P}^n$. The linear system $\mathfrak{d}$ corresponds to an invertible sheaf $\mathcal{O}_X(D)$ on X and a sub-vector space $V \subseteq \Gamma(X, \mathcal{O}_X(D))$, where $\mathfrak{d} = \mathbf{P}(V)$. By (Ex. 1.5), we have $\dim \mathfrak{d} \leq \dim |D| \leq \deg D$, so we can assume $1 \leq n \leq 4$.

If $n = 2$, then the degree genus formula gives

$$g = \frac{(d-1)(d-2)}{2} = 3.$$

If $n = 3$, then $g < 3$ by (III, Ex. 9.11) (Ex. 3.5b). We cannot have $g = 2$ by (Ex. 3.1), so $g = 0, 1$, and X is the rational quartic curve in $\mathbf{P}^3$ if $g = 0$. If $n = 4$, X is rational by (Ex. 1.5).

- (b) Let $\mathcal{I}$ be the ideal sheaf of X in $\mathbf{P}^3$, and consider the exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbf{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbf{P}^3}(2)$ and taking cohomology gives

$$0 \longrightarrow H^0(\mathbf{P}^3, \mathcal{I}(2)) \longrightarrow H^0(\mathbf{P}^3, \mathcal{O}_{\mathbf{P}^3}(2)) \longrightarrow H^0(X, \mathcal{O}_X(2)) \longrightarrow \dots$$

Now, X is an elliptic curve, and $\mathcal{O}_X(2)$ corresponds to a divisor D of degree 8. Then, Riemann Roch gives $\dim H^0(X, \mathcal{O}_X(2)) = 8$, and the vector space in the middle has dimension 10, so we conclude that $h^0(\mathbf{P}^3, \mathcal{I}(2)) \geq 2$. An element of $H^0(\mathbf{P}^3, \mathcal{I}(2))$ is a form of degree 2, whose zero-set will be a surface $Q \subseteq \mathbf{P}^3$ of degree 2 containing X . It must be irreducible and reduced because X is not contained in any $\mathbf{P}^2$, so X must be contained in two distinct irreducible quadric surfaces Q, Q' , and $Q \cap Q' = X$ because X has degree 4 (I, 7.7). $\square$

9. Prove the following lemma of Bertini: if X is a curve of degree d in $\mathbf{P}^3$, not contained in any plane, then for almost all planes $H \subseteq \mathbf{P}^3$ (meaning a Zariski open subset of the dual projective space $(\mathbf{P}^3)^*$), the intersection $X \cap H$ consists of exactly d distinct points, no three of which are collinear.

Proof. Let D be a very ample divisor of degree d corresponding to the embedding $X \hookrightarrow \mathbf{P}^3$. Bertini's theorem (II, 8.18) says for all hyperplanes H in $|D|$, the intersection $X \cap H$ consists of exactly d distinct points. $\square$

12. For each value of $d = 2, 3, 4, 5$ and r satisfying $0 \leq r \leq \frac{1}{2}(d-1)(d-2)$, show that there exists an irreducible plane curve of degree d with r nodes and no other singularities.

Proof. There is nothing to prove for $d = 2$ or $r = 0$, so assume otherwise. $\square$